LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose of Engine Vibration monitoring systems and Operation of Engine Vibration monitoring systems
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose of Engine Vibration monitoring systems and Operation of Engine Vibration monitoring systems.
5. Specific Objectives	20.5.1.1 Trainee Will Be able to explain principle of on engine Vibration monitoring systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Engine Vibration Monitoring System (EVMS) schematic diagrams and identification of sensor locations on engine cutaway models.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary cause of vibration in a rotating aircraft engine?
(b) Why is monitoring engine vibration critical for flight safety?
(c) Can you identify the main components of a typical vibration monitoring system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of engine vibration generation • Types of vibration sensors used in engines • Signal processing and data transmission • Purpose of vibration monitoring systems • Interpretation of vibration data trends	L & Ds	PPT, IB	15 min
(b)	• Operation of on-engine vibration monitoring systems • System integration with aircraft warning systems • Maintenance requirements for vibration sensors • Calibration procedures for vibration monitoring units • Troubleshooting common system failures	L & Ds	PPT, PDF	20 min
(c)	• Review of EVMS schematic diagrams • Identification of sensor locations on engine models • Analysis of vibration data trends • Discussion on maintenance schedules • Summary of key operational principles	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the principle of operation of an on-engine vibration monitoring system.
(b) Describe the steps involved in troubleshooting a vibration sensor failure.

12. Home Assignment and Follow Up	Prepare a brief report on the integration of Engine Vibration Monitoring Systems with the Engine Indicating and Crew Alerting System (EICAS) for the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to EFIS, display units, Symbol Generators and display presentations. Visit to P8i Lab for familiarization on EFIS.	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Instructor, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min